



COURSE COMPANION · BCA/B.TECH
SEMESTER I

AN ILLUSTRATED FOUNDATION COURSE



Computer Networks

From cables and clients to the OSI model, IP addressing, and wireless design — the concepts that connect the modern world, explained clearly and built up step by step.

10 Chapters

40+ Diagrams

200+ Practice Questions

Compiled & Restructured Edition

First Edition · 2026
Study & Reference Copy

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Preface

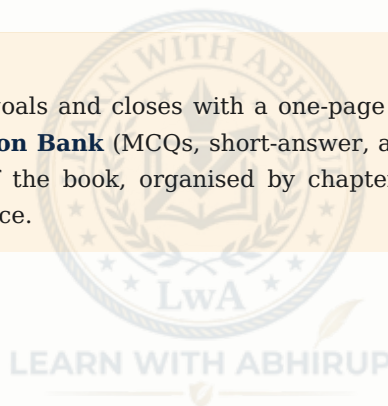
Every device you own — the phone in your pocket, the laptop on your desk, the smart speaker in your kitchen — is a node in a vast, layered system of wires, radio waves, addresses, and agreements about how to talk to one another. That system is what this book calls a **computer network**, and understanding it is one of the most practically useful things a student of technology can do, because almost nothing in modern computing works without it.

This book began as a semester-one study manual written for first-year students. That original material was accurate and well organised, but it read like a set of lecture notes — dense paragraphs, few visuals, and a question bank bolted on at the end. This edition keeps every fact from that source, checks it, corrects the handful of errors that had crept in (notably a subnetting worked example that used the wrong default mask — see Chapter 7), and rebuilds it as an actual textbook: chapters with a clear narrative arc, original diagrams for every major concept, worked examples, "common mistake" warnings for the things students reliably get wrong on exams, and a review system at the end of every chapter.

The ten chapters move in the order a network actually gets built: first you learn what a network is and how its pieces are shaped (Chapters 1-2), then how it is physically wired together (Chapters 3-4), then the rulebooks that let wildly different devices understand each other (Chapters 5-6), then the addressing system that gives every device a "postal address" (Chapter 7), then the two dominant families of implementation — wired Ethernet and wireless (Chapters 8-9) — and finally the security discipline that has to sit on top of all of it (Chapter 10).

HOW TO USE THIS BOOK

Each chapter opens with learning goals and closes with a one-page summary and a "Check Yourself" question set. A consolidated **Question Bank** (MCQs, short-answer, and long-answer questions with an answer key) appears at the end of the book, organised by chapter, for exam revision. A **Glossary** defines every bolded term in one place.



CHAPTER 1

What Is a Computer Network?

The people, machines, and rules that make communication possible — and why almost everything you do on a device depends on them.

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- Define a computer network and identify its four core components
- Explain why organisations build networks instead of using standalone computers
- Classify a network by geographic scale: PAN, LAN, CAN, MAN, WAN
- Distinguish a Storage Area Network and a Virtual Private Network from the above

1.1 Defining the Network

A **computer network** is a collection of interconnected computers and other devices that can communicate with one another and share resources. The devices involved might sit in the same room or on opposite sides of the planet — geography does not change the definition, only the technology used to bridge the distance. What matters is that two conditions are met simultaneously: devices are *connected* by some medium, and they are able to *exchange meaningful data* over that connection.

KEY TERM · NODE

Any device attached to a network — a computer, printer, phone, router, or server — is called a **node**. A network, at its simplest, is just a set of nodes plus the links between them.

1.2 The Four Building Blocks

Strip any network down to its essentials and you find the same four ingredients every time:

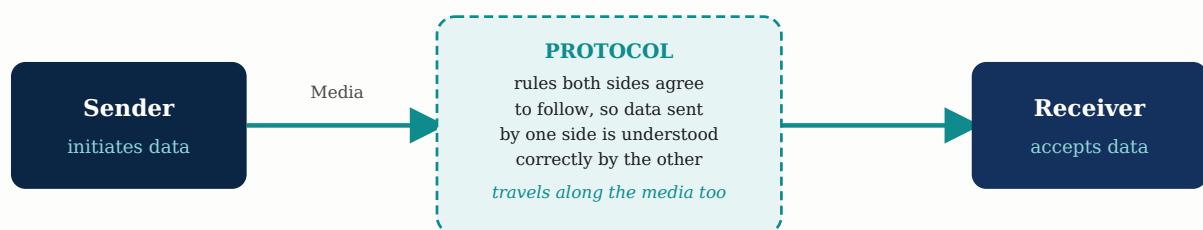


Figure 1.1 — The four components present in every network exchange: a sender, a receiver, the media carrying the signal, and the protocol governing how it is interpreted.

Component	Role
Sender	The device or entity that originates the data — a computer, phone, server, or sensor.
Receiver	The device or entity that accepts and interprets the data sent by the sender.
Media	The physical or wireless channel data travels across — twisted-pair copper, coaxial cable, optical fibre, radio waves, microwaves, or infrared light.
Protocol	The agreed-upon set of rules governing how data is formatted, transmitted, and interpreted, so that sender and receiver — even from different manufacturers — understand each other.

1.3 Why Build a Network at All?

It would be technically possible to run every computer in a building as an island, moving files around on USB drives. Networks exist because that approach does not scale. Once you connect machines together, a long list of practical benefits follow immediately:

RESOURCE SHARING

Printers, storage, and databases can serve many users instead of needing a duplicate for each desk.

COMMUNICATION

Email, messaging, and video calls become possible in real time regardless of physical location.

CENTRALISED MANAGEMENT

Large organisations administer security, backups, and software updates from one place.

SCALABILITY

New devices and users join without redesigning the whole system.

REDUNDANCY

Multiple paths between devices mean one failed link does not have to mean a failed connection.

ACCESS TO REMOTE RESOURCES

Cloud storage and computing depend entirely on network reach.

1.4 Classifying Networks by Geographic Scale

The most common way to categorise a network is by how much physical territory it covers. Scale changes almost everything about a network's design — its cost, the technology it relies on, and who manages it.

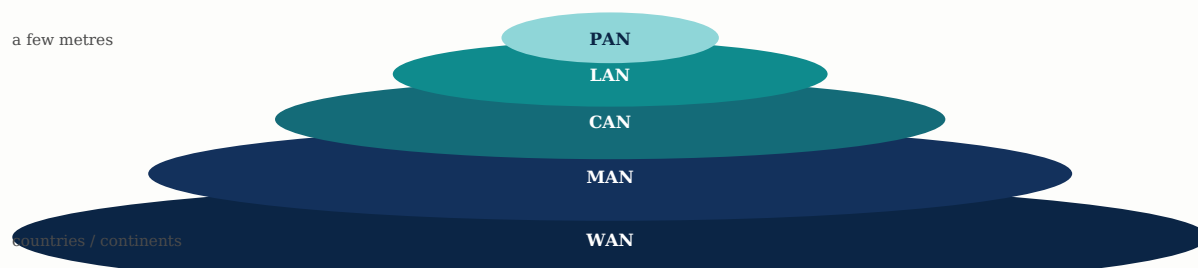


Figure 1.2 — Network types nest by geographic scale, from a personal area network around one person out to a wide area network spanning continents.

Personal Area Network (PAN)

Covers the space within a person's immediate reach and links personal devices — a phone, a laptop, wireless earbuds — typically over Bluetooth or infrared.

Local Area Network (LAN)

Confined to a single building or office. LANs are inexpensive to build and maintain and typically use Ethernet cabling or Wi-Fi to connect computers, printers, and shared storage.

Campus Area Network (CAN)

A specialised, larger LAN that connects several buildings within one university, college, or corporate campus, using the same high-speed technologies as a MAN but under single-site ownership.

Metropolitan Area Network (MAN)

Spans a city or a large campus and stitches multiple LANs together, usually over fibre-optic cabling, to give high-bandwidth transfer across a metropolitan region.

Wide Area Network (WAN)

Covers countries or continents, linking geographically distant LANs and MANs over leased lines, satellite links, or microwave relays. The internet is the largest WAN in existence.

ALSO WORTH KNOWING

Virtual Private Network (VPN): not a scale category but a security technique — it creates an encrypted tunnel across a public network (usually the internet) so a remote user can reach a private network as if they were plugged into it directly.

Storage Area Network (SAN): a high-speed, dedicated network that connects storage devices to servers, giving centralised, fast data storage separate from the general-purpose LAN.

COMMON MISTAKE

Students often treat "MAN" and "CAN" as interchangeable. A CAN is owned and managed by a single institution (a university, a corporate park); a MAN typically spans infrastructure owned by a telecom provider serving an entire city, including multiple unrelated organisations.

Chapter Summary

- A network is any set of interconnected devices (nodes) that can exchange data — built from a sender, a receiver, transmission media, and a shared protocol.
- Networks exist because they enable resource sharing, real-time communication, centralised management, scalability, redundancy, and remote access.
- Networks are classified by geographic reach: PAN < LAN < CAN < MAN < WAN.
- VPNs and SANs solve specific problems (secure remote access, centralised storage) rather than describing a size category.

Check Yourself

1. Name the four components present in any data exchange between two devices.
2. Your college wants to connect three buildings on one campus using fibre. Is this a CAN or a MAN, and why?
3. Why is a VPN not considered a fifth entry in the PAN/LAN/CAN/MAN/WAN scale?
4. List three business benefits of networking that would be lost if every computer worked in isolation.



CHAPTER 2

Network Architectures and Topologies

Two questions every network designer must answer: who is in charge, and what shape does the wiring take?

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- Contrast peer-to-peer and client-server architectures
- Distinguish physical topology from logical topology
- Compare bus, star, ring, mesh, tree, and hybrid topologies on cost, reliability, and scalability
- Recommend a topology for a given scenario

2.1 Who Is in Charge? Peer-to-Peer vs. Client-Server

Before worrying about cables, a designer must decide how responsibility is distributed among the machines on the network.

Peer-to-peer

In a **peer-to-peer (P2P)** network, every node has equal standing: there is no dedicated central server, and each computer can act as both a client (requesting services) and a server (providing them) at the same time. Each machine stores its own data and manages its own security independently.

Client-server

In a **client-server** network, files and services are centralised on one or more dedicated, powerful machines — the **servers** — while ordinary user machines, the **clients**, request services from them. On a large network there may be many servers, each specialised: a file server, a mail server, a print server, and so on.

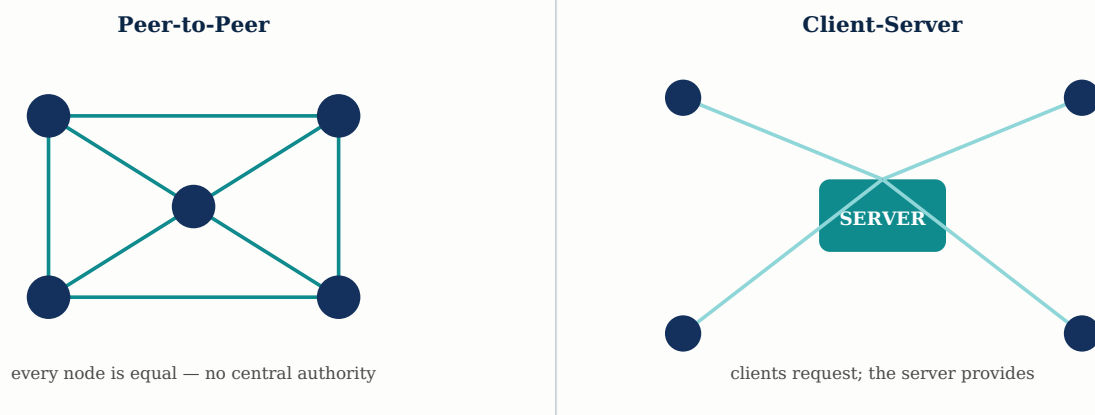


Figure 2.1 — Peer-to-peer distributes responsibility evenly; client-server concentrates it in dedicated machines.

Aspect	Peer-to-Peer	Client-Server
Central authority	None	One or more dedicated servers
Cost	Low — no dedicated server hardware	Higher — servers must be purchased and maintained
Security management	Handled separately on each node	Centralised, easier to enforce consistently
Best suited for	Small offices, home networks (2-10 devices)	Organisations needing shared, centrally controlled resources

2.2 Physical vs. Logical Topology

Topology describes the arrangement of nodes and links in a network. It has two faces that are frequently confused:

- **Physical topology** — the actual, visible layout of cables and devices.
- **Logical topology** — the path data actually follows, which can differ from the physical layout. (A network can be physically wired as a star, using a central switch, while data logically behaves as if it were on a shared bus.)

2.3 The Six Standard Topologies

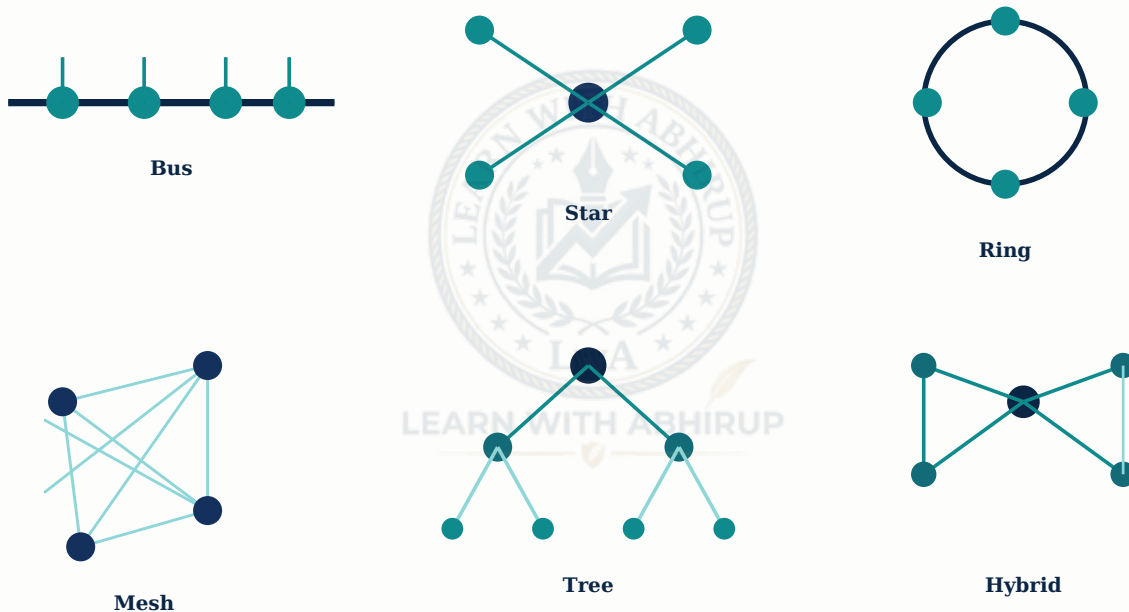


Figure 2.2 — The six standard network topologies. Dots represent nodes; lines represent physical links.

Topology	How it works	Strength	Weakness
Bus	All devices tap into one shared central cable	Cheap, simple to install	One cable break disables the whole network; collisions increase with more devices
Star	Every device connects to a central hub/switch	Easy to install and troubleshoot; one device failing doesn't affect others	Central device is a single point of failure; needs more cabling
Ring	Devices form a closed loop; data travels one direction	Orderly, predictable transmission — good for real-time traffic	One broken node can disrupt the whole ring; hard to reconfigure
Mesh	Devices have multiple redundant paths to each other	Extremely fault tolerant and scalable	Expensive; complex to install and manage
Tree	Hierarchical — a root switch branches into star sub-networks	Organised, scalable structure for large sites	Failure high in the hierarchy can disrupt many branches at once
Hybrid	Combines two or more of the above	Flexible, tuned to the actual requirement	Design and management complexity of whichever topologies are mixed

WORKED EXAMPLE

A college is wiring three departmental buildings and wants (a) departments to keep working if one building's internal network fails, and (b) a controllable, budget-friendly design.

Recommended design: a star topology *inside* each building (cheap, easy to troubleshoot, isolates faults to one device) connected through a tree/hybrid backbone between buildings — this is exactly how most real campus networks are built, and is why the "hybrid" topology is the one most large organisations actually use in practice.

Chapter Summary

- Architecture (who is in charge) and topology (what shape the wiring takes) are two separate design decisions.
- Peer-to-peer suits small, informal networks; client-server suits organisations needing centralised control.
- Physical topology is the visible cabling; logical topology is the path data actually follows — they can differ.
- Star is the default choice for most modern LANs; mesh is chosen when fault tolerance outweighs cost; hybrid designs dominate in large real-world networks.

Check Yourself

1. A small design studio with five employees wants the cheapest possible file-sharing setup. Would you recommend peer-to-peer or client-server, and why?
2. Explain, with an example, how a network can be physically a star but logically behave like a bus.
3. Which topology would you pick for a hospital's life-critical monitoring network, and what property of that topology justifies the choice?
4. Why is "hybrid" almost always the honest answer to "what topology does your organisation use"?

CHAPTER 3

The Internet and Its Applications

How billions of independent networks agree to act as one, and what we actually build on top of that agreement.

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- Describe the internet as a "network of networks" and name its core building blocks
- Trace what happens, at a conceptual level, when a device requests a web page
- List major categories of network applications and one example of each

3.1 The Internet: A Network of Networks

The **internet** is not a single network — it is a global mesh of interconnected networks, all agreeing to use the same addressing scheme and the same core protocols so that data can cross from one network to the next without anyone in the middle needing to understand its content. Four ingredients make this possible:

Devices and endpoints

Every participating device — computer, phone, server, router — is given a unique **IP address** that functions as its identity on the network (Chapter 7 covers this in depth).

Internet Service Providers (ISPs)

ISPs own or lease the physical infrastructure — fibre, cable, satellite links — that connects individual users and organisations to the wider internet backbone.

Routers and switches

Routers examine the destination address of each packet of data and forward it toward that destination, hopping across networks; switches move data efficiently within a single network (both are covered fully in Chapter 4).

Protocols

A shared rulebook — chiefly **TCP/IP** — governs how data is broken into packets, addressed, routed, and reassembled correctly at the far end, regardless of which hardware vendors or operating systems are involved on either side.

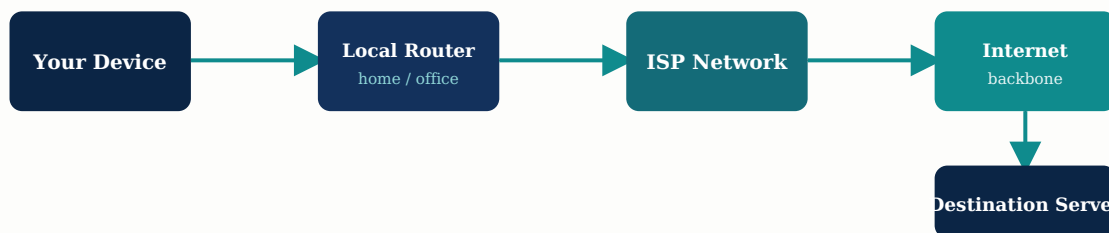


Figure 3.1 — A simplified path from your device to a destination server: local router → ISP → internet backbone → destination network.

THE SHORT VERSION

When you open a web page, your device asks a router to forward a request; the router passes it to your ISP; the ISP routes it across the internet backbone, hopping between networks, until it reaches the network hosting the destination server; the server's reply retraces a path (not necessarily the same one) back to you — typically within a few hundred milliseconds.

3.2 Categories of Network Applications

Category	What it does	Everyday examples
Communication	Real-time or asynchronous exchange of messages between people	Email, instant messaging, video conferencing
File sharing	Uploading, downloading, and collaborating on files stored remotely	Cloud drives, FTP, shared network folders
Online services	Content and interaction delivered entirely over the network	Web browsing, streaming media, online gaming, social media

Email in a little more detail

Email remains one of the oldest and most heavily used network applications. A message is composed on a client, handed to a mail server which routes it (often through several intermediate servers) to the recipient's mail server, where it waits until the recipient's client retrieves it. The whole system relies on a handful of protocols — SMTP for sending, and POP3 or IMAP for retrieving — running on top of the same TCP/IP foundation that carries every other kind of internet traffic.

Chapter Summary

- The internet is a "network of networks" held together by universal addressing (IP addresses) and a shared protocol suite (TCP/IP).
- A request typically travels device → local router → ISP → backbone → destination network, and the reply need not retrace the same path.
- Network applications fall into three broad categories: communication, file sharing, and online services.

Check Yourself

1. Why is the internet described as a "network of networks" rather than a single network?
2. Name the two protocols responsible for, respectively, sending and retrieving email.
3. Which single component in Figure 3.1 would you say is most responsible for making your private request "public" — i.e., routable to a destination you've never directly connected to?



CHAPTER 4

Network Devices

The hardware that connects, directs, and protects traffic — from the network card in your laptop to the firewall guarding the front door.

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- Explain the role of a NIC and compare its wired, wireless, USB, and fibre variants
- Differentiate hubs, switches, routers, bridges, gateways, and modems by OSI layer and function
- Explain what a MAC address is and how it differs from an IP address
- Classify the major types of firewalls

4.1 The Network Interface Card (NIC)

A **Network Interface Card** is the hardware component that gives a device its physical connection to a network. It converts data from the parallel form used inside a computer to the serial form needed for transmission, and it carries a globally unique **MAC address** that identifies the device on its local network.

NIC Type	Connects via	Advantage	Disadvantage
Wired	Ethernet cable	Fast, stable, resistant to interference	Limited mobility; cabling can be cumbersome
Wireless	Radio waves	Mobility and flexible placement	Susceptible to interference; typically slower than wired
USB	USB port	Portable, plug-and-play	Lower performance; more prone to physical damage
Fibre optic	Optical fibre cable	Very high speed over long distances	Expensive; specialised installation skills required

KEY TERM · MAC ADDRESS

A **MAC (Media Access Control) address** is a 12-digit hexadecimal number burned into a NIC by its manufacturer, written as pairs like `00:1A:2B:3C:4D:5E`. The first six digits (the OUI) identify the manufacturer; the last six uniquely identify that specific card. Unlike an IP address, a MAC address is permanent hardware identity, not a network-assigned "location" — which is why **ARP (Address Resolution Protocol)** exists, specifically to map a device's IP address to its MAC address on the local network.

4.2 Hubs, Switches, and the Layer They Operate At

Both hubs and switches connect multiple devices in a LAN, but they behave very differently.

HUB · OSI LAYER 1

Broadcasts every incoming signal out of every other port, regardless of the intended recipient. Cheap

and simple, but wasteful — it causes unnecessary traffic and collisions as more devices are added.

SWITCH · OSI LAYER 2

Reads the destination MAC address of each frame and forwards it only to the port the recipient is on. This cuts congestion dramatically and allows full-duplex (simultaneous send/receive) communication.

Types of switches

- **Unmanaged switch** — plug-and-play, no configuration options.
- **Managed switch** — configurable monitoring, traffic control, and access rules; used where administrators need real oversight.
- **Smart switch** — a lighter-weight managed switch with a simple web interface for mid-sized networks.
- **KVM switch** — lets one keyboard/video/mouse console control several computers.
- **PoE switch** — delivers electrical power and data over the same Ethernet cable, commonly used for IP cameras and Wi-Fi access points.

4.3 Routers, Bridges, and Gateways

Device	OSI Layer	Function
Router	3 (Network)	Connects different networks and forwards IP packets between them using a routing table, choosing the most efficient path. Provides some protection against broadcast storms by not forwarding local broadcasts onward.
Bridge	2 (Data Link)	Joins two network segments so they behave as one, filtering traffic by MAC address to reduce unnecessary crossover.
Gateway	Varies (often 7)	Sits between two dissimilar networks or protocol systems and translates between them — a router may perform some gateway functions.

COMMON MISTAKE

Students often use "router" and "gateway" interchangeably. A router's defining job is forwarding IP packets between networks using addresses; a gateway's defining job is *protocol translation* between systems that don't natively speak the same language. In home networking the same physical box usually does both jobs, which is where the confusion comes from.

4.4 Modems

A **modem** (MODulator-DEModulator) converts the digital signals used inside a computer into the analogue signals needed to travel across telephone or cable infrastructure, and back again on receipt. Modern broadband and cable modems are far faster than the dial-up modems the term originally described, but the core job — converting between digital and analogue — is unchanged.

4.5 Firewalls

A **firewall** is a security device or piece of software that monitors and filters network traffic against a defined policy, sitting conceptually between a trusted private network and an untrusted public one.

Firewall Type	How it works
Packet filtering	Examines each packet against fixed rules at the network layer (Layer 3) — fast, but relatively unsophisticated.
Proxy / application-layer	Sits between client and server, hiding the client's real address and inspecting traffic all the way up at Layer 7.
IDS / IPS	Not firewalls in the strict sense, but companion systems: an Intrusion Detection System watches for malicious patterns; an Intrusion Prevention System actively blocks them.
Hardware firewall	A dedicated physical appliance protecting an entire network.
Software firewall	Installed on an individual device to protect just that device.
Cloud firewall	Deployed within cloud infrastructure to protect cloud-hosted resources.

Chapter Summary

- A NIC provides a device's physical network connection and carries its unique MAC address.
- Hubs (Layer 1) broadcast blindly; switches (Layer 2) forward intelligently by MAC address.
- Routers (Layer 3) connect different networks using IP addresses; bridges join segments at Layer 2; gateways translate between dissimilar protocol systems.
- Modems convert between digital and analogue signals for transmission over telephone/cable lines.
- Firewalls range from simple packet filters to application-aware proxies, and exist as hardware, software, and cloud-native deployments.

Check Yourself

1. A small office notices its network slows down badly as more computers are added to its hub. What device would you recommend as a replacement, and why does it solve the problem?
2. What is the essential difference between a router and a gateway?
3. Why can't a MAC address be used, by itself, to route traffic across the internet the way an IP address can? (Hint: think about scale.)
4. Which firewall type would best protect a single laptop used on untrusted public Wi-Fi?

CHAPTER 5

Transmission Media and Connectors

Copper, glass, and air — the three physical substances that carry every bit you've ever sent.

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- Compare coaxial, twisted-pair, and fibre-optic cabling
- Match each cable family to its standard connectors
- Explain why RJ45 and RJ11 are not interchangeable

5.1 Coaxial Cable

Coaxial cable carries a signal down a central copper conductor, shielded by an insulating layer and a woven metal shield — a design that resists interference well, historically making it the backbone of cable television and early Ethernet networks.

Type	Typical use
RG-6	Cable TV and internet — thicker insulation, better shielding than RG-59
RG-59	Low-frequency use such as CCTV and analogue video
RG-11	Long-distance, high-frequency runs — lower signal loss (attenuation)

Connectors: **BNC** (bayonet-locking, professional video/RF), **F-Type** (threaded, cable TV/internet), **N-Type** (weatherproof, outdoor antennas and satellite dishes).

5.2 Twisted-Pair Cable

Twisted-pair cable is the dominant medium for modern Ethernet LANs. Pairs of copper wires are twisted together specifically to cancel out electromagnetic interference between them.

- **UTP (Unshielded Twisted Pair)** — no extra shielding, flexible, easy to install; the standard choice for office and home networks.
- **STP (Shielded Twisted Pair)** — adds shielding for environments with heavy electromagnetic interference.

Category	Max Speed	Max Frequency
Cat 5e	1 Gbps	100 MHz
Cat 6	10 Gbps (limited distance)	250 MHz
Cat 6a	10 Gbps	500 MHz

RJ45 VS. RJ11 — DON'T MIX THEM UP

Key	RJ45	RJ11
Used for	Ethernet networking	Telephone lines
Size	Larger	Smaller
Wires	8 (8P8C)	4 (6P4C)
Bandwidth	Up to 10 Gbps	Up to 24 Mbps

5.3 Fibre-Optic Cable

Fibre carries data as pulses of light through a glass or plastic core, immune to electromagnetic interference and capable of very high speeds over very long distances.

- **Single-Mode Fibre (SMF)** — a narrow core (~9 microns) carrying one light path; used for long-distance, high-bandwidth links (telecom backbones, city-to-city runs).
- **Multi-Mode Fibre (MMF)** — a wider core (50 or 62.5 microns) carrying multiple light paths; cheaper, used for shorter runs within buildings or data centres.

Connectors: **SC** (push-pull, square, easy to use), **LC** (compact, ideal for high-density racks), **ST** (bayonet-style, quick secure connections).



Figure 5.1 — The three cable families and the connectors most commonly paired with each.

Chapter Summary

- Coaxial cable (RG-6/RG-59/RG-11) is largely legacy in LANs but still common for cable TV/internet drops.
- Twisted-pair cable (UTP/STP, Cat 5e-6a) is the standard medium for modern wired LANs, terminated in RJ45.
- Fibre optic cable (single-mode for distance, multi-mode for cost-effective short runs) offers the highest speed and immunity to electromagnetic interference.
- Connector families are not interchangeable across cable types — matching connector to cable is a basic but frequently tested skill.

Check Yourself

1. Why does twisting the wire pairs in twisted-pair cable actually help signal quality?
2. You need to run a network connection 40 km between two campus buildings. Which cable family would you choose, and which sub-type?
3. A colleague plugs a telephone cable into an Ethernet switch port and it doesn't fit. Explain why, referencing wire count and connector type.

CHAPTER 6

Reference Models: OSI and TCP/IP

A universal vocabulary for describing what happens to your data, one layer at a time, between the moment you hit "send" and the moment it arrives.

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- List the seven OSI layers in order and state one function of each
- List the four TCP/IP layers and map them onto the OSI model
- Explain why these models make network troubleshooting more systematic

6.1 Why We Need a Layered Model

A modern network connection involves cabling, electrical signalling, addressing, routing, reliability, and application-level meaning — all happening at once, built by different vendors running different software. A **layered model** breaks this into independent stages, each with a clearly defined job, so that a change at one layer (say, swapping Wi-Fi for Ethernet) doesn't require rebuilding every layer above it.

6.2 The OSI Model

The **OSI (Open Systems Interconnection) model**, developed by the International Organization for Standardization (ISO), divides network communication into seven layers.

7	Application	HTTP, FTP, SMTP — user-facing services
6	Presentation	Encryption, compression, formatting
5	Session	Session set-up / teardown
4	Transport	TCP, UDP — reliable delivery, segmentation
3	Network	IP addressing and routing
2	Data Link	MAC addresses, frames, error checking
1	Physical	Bits, cables, voltage, connectors

Mnemonic: "All People Seem To Need Data Processing" (top → bottom)

Figure 6.1 — The seven OSI layers, application at top (closest to the user) down to physical at the bottom (closest to the wire).

#	Layer	Job	Examples
7	Application	Provides network services directly to user-facing software	HTTP, FTP, SMTP
6	Presentation	Formats, encrypts/decrypts, and compresses data for the application layer	Encryption, character-set conversion
5	Session	Establishes, manages, and terminates communication sessions	Session initiation and teardown
4	Transport	Provides reliable, ordered, error-checked delivery between applications	TCP, UDP
3	Network	Routes packets between networks using logical (IP) addresses	IP routing, packet forwarding
2	Data Link	Ensures error-free transmission of frames on the same physical link	MAC addresses, error checking
1	Physical	Transmits raw bits across a physical or wireless medium	Connectors, cables, voltage levels

6.3 The TCP/IP Model

While OSI is the conceptual reference every course teaches, the internet actually runs on the simpler, four-layer **TCP/IP model**.

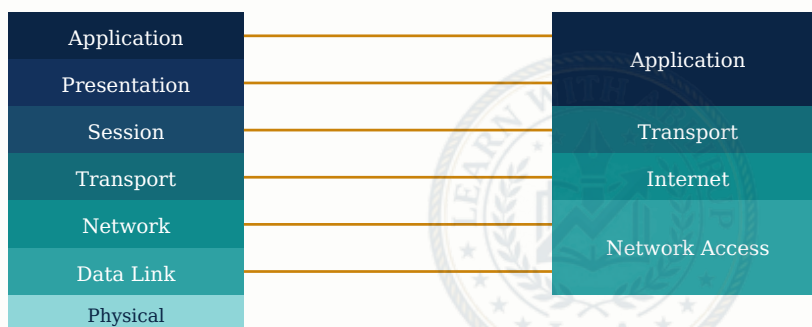


Figure 6.2 — Mapping the seven OSI layers onto the four TCP/IP layers: TCP/IP merges Application+Presentation+Session into one Application layer, and Data Link+Physical into one Network Access layer.

TCP/IP Layer	Combines OSI layers	Job
Application	7, 6, 5	Provides services like HTTP, FTP, and email directly to user applications
Transport	4	Reliable data transfer via TCP or UDP
Internet	3	Routes packets across networks based on IP addresses
Network Access	2, 1	Physical transmission and MAC-level media access control

6.4 OSI vs. TCP/IP at a Glance

OSI Model	TCP/IP Model
Developed by ISO (International Organization for Standardization)	Developed for ARPANET (Advanced Research Projects Agency Network)
7 layers	4 layers
A conceptual reference — protocols underneath it can be swapped out	A working implementation — the actual protocol stack running the internet today
Mostly used for teaching and troubleshooting terminology	What is actually deployed everywhere

WHY BOTHER WITH TWO MODELS?

OSI gives you a precise, universal vocabulary — "that's a Layer 2 problem," "this needs a Layer 7 firewall" — that troubleshooters and textbooks rely on worldwide. TCP/IP is what your computer is actually running. Learning both means you can talk about networks precisely *and* understand what's happening on the wire.

Chapter Summary

- The OSI model's seven layers (Physical, Data Link, Network, Transport, Session, Presentation, Application) give a universal vocabulary for network functions.
- The TCP/IP model's four layers (Network Access, Internet, Transport, Application) describe what is actually implemented across the internet.
- Each TCP/IP layer maps onto one or more OSI layers; both models describe the same reality at different resolutions.

Check Yourself

1. Which OSI layer is responsible for choosing the path a packet takes across multiple networks?
2. A network administrator says "this is a Layer 2 issue." What kind of device and addressing scheme should they be investigating?
3. Name the three OSI layers that TCP/IP merges into its single Application layer.
4. Why does swapping a LAN's Ethernet cabling for Wi-Fi not require any change to how a browser requests a web page?

CHAPTER 7

IP Addressing, NAT, and Subnetting

Every device's postal address, how millions of private devices share one public one, and how to carve a network into smaller, well-organised pieces.

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- Describe the structure and classes of an IPv4 address
- Distinguish unicast, broadcast, multicast, public, and private addresses
- Explain what NAT does and why it matters
- Perform basic subnetting calculations, including CIDR notation

7.1 What an IPv4 Address Is

An **IPv4 address** is a 32-bit numeric label that uniquely identifies a device on a network. It is almost always written in **dotted-decimal notation**: four 8-bit sections ("octets"), each ranging from 0 to 255, separated by periods.

BINARY TO DECIMAL

11000000.10101000.00000000.00000001 in binary is the same address as 192.168.0.1 in decimal — each block of 8 bits converts independently into one octet.

Every IPv4 address has two logical parts:

- **Network portion** — identifies which network the device belongs to.
- **Host portion** — identifies the specific device within that network.

Where the boundary between the two falls is determined entirely by the **subnet mask** (Section 7.4) — the same 32-bit address can belong to a small subnet or a huge one depending on where that boundary is drawn.

7.2 The Five Address Classes

Class	Range	Typical use
A	0.0.0.0 - 127.255.255.255	Very large networks (few networks, huge number of hosts each)
B	128.0.0.0 - 191.255.255.255	Medium-sized networks
C	192.0.0.0 - 223.255.255.255	Small networks (many networks, fewer hosts each)
D	224.0.0.0 - 239.255.255.255	Reserved for multicast groups
E	240.0.0.0 - 255.255.255.255	Reserved, experimental / future use

7.3 Special and Reserved Addresses

Type	Range / Example	Purpose
Loopback	127.0.0.1	Testing and troubleshooting on the local machine itself
Broadcast	255.255.255.255 (or the subnet's broadcast address, e.g. 192.168.1.255)	Sends data to every device on a network segment at once
Private (Class A)	10.0.0.0 - 10.255.255.255	Internal use, not routable on the public internet
Private (Class B)	172.16.0.0 - 172.31.255.255	Internal use
Private (Class C)	192.168.0.0 - 192.168.255.255	Internal use
APIPA	169.254.0.0 - 169.254.255.255	Self-assigned automatically when a device can't reach a DHCP server

UNICAST

One sender, one specific recipient — the default mode for almost all everyday traffic, such as a client talking to a server.

BROADCAST

One sender, every device on the local segment — used for things like ARP requests or DHCP discovery.

MULTICAST (224.0.0.0-239.255.255.255)

One sender, a specific group of interested recipients — used for streaming media and online gaming.

PUBLIC VS. PRIVATE

Public addresses are globally unique and routable on the internet, allocated by IANA and regional registries; private addresses are reusable inside any organisation's own network and never routed on the public internet directly.

7.4 Subnet Masks and CIDR Notation

A **subnet mask** is a 32-bit pattern of contiguous 1s followed by 0s that marks where the network portion of an address ends and the host portion begins. For example, the mask `255.255.255.0` means the first three octets are the network portion and the last octet is available for host addresses.

CIDR (Classless Inter-Domain Routing) notation writes the same information more compactly: an address followed by a slash and the number of leading 1-bits in the mask. `192.168.1.0/24` means "the network 192.168.1.0, with a 24-bit mask" — identical information to writing out `255.255.255.0` separately.

SUBNETTING, IN ONE SENTENCE

Subnetting is the technique of borrowing bits from the host portion of an address to create multiple smaller networks (subnets) out of one larger one — trading host capacity for organisation, security isolation, and traffic control.

7.5 Worked Subnetting Problems

PROBLEM 1

An address `156.36.2.58` uses a default mask that has been extended to `/22`. Express this subnet mask in dotted-decimal form.

Solution: `/22` means the first 22 bits of the mask are 1s:

`11111111.11111111.11111100.00000000`

Converting each octet: 255, 255, 252, 0 → the subnet mask is `255.255.252.0`.

PROBLEM 2

Given the address `192.168.1.0` with subnet mask `255.255.255.128`, find the network address, broadcast address, and usable host range.

Solution: The mask's last octet, 128, is `10000000` in binary — one extra network bit is borrowed from the last octet, splitting it into two subnets of 128 addresses each: one starting at `.0` and one starting at `.128`.

Since the starting address is `192.168.1.0`, it falls in the first block:

- **Network address:** `192.168.1.0`
- **Broadcast address:** `192.168.1.127` (the last address before the next block begins)
- **Usable host range:** `192.168.1.1` – `192.168.1.126`

PROBLEM 3

The network `172.16.0.0` is given CIDR notation `/20`. Find the number of subnets created and the number of usable hosts per subnet.

Solution: `172.16.0.0` is a Class B address, whose *default* mask is `/16`. Extending the mask to `/20` borrows $20 - 16 = 4$ bits for subnetting, and leaves $32 - 20 = 12$ bits for hosts.

- **Number of subnets:** $2^4 = 16$ subnets
- **Usable hosts per subnet:** $2^{12} - 2 = 4,094$ hosts (subtracting the network address and the broadcast address from each subnet)

COMMON MISTAKE — AND A CORRECTION

It's tempting to count subnet and host bits from the `/24` boundary out of habit, since so many worked examples default to Class C. Here, `172.16.0.0` is a **Class B** address with a **/16** default mask, not a `/24` — so the bits borrowed for subnetting are counted from 16, not 24. Getting this starting point wrong is the single most common subnetting error in exams, and it silently produces numbers that look plausible (an answer of "8 subnets, 30 hosts") but are actually wrong once you check which class the address belongs to.

7.6 Network Address Translation (NAT)

NAT lets many devices carrying private IP addresses share one public IP address when talking to the internet. Picture a household with a dozen devices, all using addresses in the `192.168.1.x` range: when any of them requests a web page, the router's NAT function swaps the device's private address for the router's single public address on the way out, and reverses the swap for the reply on the way back in — silently keeping track of which internal device each conversation belongs to.



Figure 7.1 — Three private addresses share a single public address; NAT keeps track of which reply belongs to which internal device.

NAT Type	Behaviour
Static NAT	Assigns one fixed public address to one specific internal device
Dynamic NAT	Allocates a public address from a pool as needed — common in homes
PAT (Port Address Translation)	Shares a single public address among many devices by distinguishing them with port numbers — the most common form of NAT today

Chapter Summary

- An IPv4 address is a 32-bit, dotted-decimal identifier split into a network portion and a host portion.
- Addresses fall into five classes (A-E); private ranges (10.x, 172.16–31.x, 192.168.x) are reserved for internal use and never routed publicly.
- Subnet masks and CIDR notation define where the network/host boundary sits; subnetting borrows host bits to create additional, smaller networks.
- Always check an address's *default* class mask before counting borrowed subnet bits — this is the step most subnetting mistakes skip.
- NAT (commonly as PAT) lets many private addresses share one public address, conserving address space and adding a layer of obscurity.

Check Yourself

1. What class does the address 172.16.5.10 belong to, and what is its default subnet mask?
2. Given the network 192.168.10.0/26, find the number of subnets available if it replaces the default /24 mask, and the number of usable hosts per subnet.
3. Explain, in your own words, why NAT is considered a "basic security layer" even though it was not originally designed as a security feature.
4. Why is 169.254.x.x on a device usually a sign that something has gone wrong, rather than correct configuration?

CHAPTER 8

LAN Technologies and the Ethernet Standard

How the IEEE turned "every vendor does it differently" into a single set of rules the entire industry follows — and how that rulebook evolved from 10 Mbps to gigabit speeds.

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- Explain the purpose of the IEEE 802 project and name its key sub-standards
- Define core LAN vocabulary: node, packet, protocol, server, terminal
- Compare Ethernet, Token Ring, Token Bus, and FDDI
- Trace Ethernet's evolution from 10Base5 through Gigabit Ethernet
- Contrast CSMA/CD and CSMA/CA

8.1 The IEEE 802 Project

By the early 1980s, computer manufacturers were building incompatible networking hardware, each with its own idea of how devices should talk to each other. In 1985 the IEEE Computer Society launched **Project 802** to define a shared set of standards so that any vendor's equipment could interoperate on a LAN or MAN.

Standard	Covers
IEEE 802	Overall LAN/MAN standards
IEEE 802.1	LAN/MAN standards and MAC bridging
IEEE 802.2	Logical Link Control (LLC)
IEEE 802.3	Ethernet (bus topology)
IEEE 802.4	Token Bus
IEEE 802.5	Token Ring
IEEE 802.11	Wireless LAN (Wi-Fi) — see Chapter 9

8.2 Core LAN Vocabulary

NODE

A connection point in the network — an endpoint for sending, receiving, or forwarding data.

SERVER

A device or system that stores and processes information for other devices on the network, ranging from a single physical box to a virtual server in the cloud.

TERMINAL

A device with little or no processing power of its own that relies entirely on a back-end host to do the work — the origin of the term "dumb terminal."

PACKET

The smallest unit of data that travels across a network at one time, containing both the payload and control information (source/destination address, sequencing, error checking) in its header and trailer.

8.3 Competing LAN Technologies

Ethernet

Created at Xerox PARC in 1973 by Robert Metcalfe and colleagues, **Ethernet** became the dominant wired LAN technology because of its simplicity: plug a cable into an Ethernet port, connect it to a switch or router, and the device is on the network.

Token Ring (IEEE 802.5)

Stations are arranged in a ring, and a small 3-byte **token** circulates around it. A station may only transmit data while it is holding the token; once its frame is sent and acknowledged, it passes the token onward. This guarantees orderly, collision-free access, at the cost of a station failure being able to disrupt the entire ring.

Token Bus (IEEE 802.4)

Applies the same token-passing discipline as Token Ring, but over a physical bus or tree topology using coaxial cable. Stations form a *virtual* ring — each one knows its predecessor and successor by address — even though the physical wiring is a bus.

FDDI (Fiber Distributed Data Interface)

An ANSI/ISO standard for high-speed LANs over fibre optic cable, capable of 100 Mbps across distances up to 200 km. FDDI uses two counter-rotating token rings — a primary ring for normal traffic and a secondary ring that takes over automatically if the primary ring fails.

Feature	802.3 (Ethernet)	802.4 (Token Bus)	802.5 (Token Ring)
Topology	Bus	Bus or tree	Ring
Priority support	None	Yes	Yes
Protocol complexity	Simple	Complex	Moderate
Real-time traffic	Not well suited	Well suited	Well suited
Behaviour under heavy load	Efficiency drops (collisions)	Stays efficient	Stays efficient

8.4 The Ethernet Family, Generation by Generation

Standard	Medium	Speed	Notes
10Base5 (Thicknet)	Thick coaxial	10 Mbps, up to 500 m	Original Ethernet cabling; obsolete
10Base2 (Thinnet)	Thin coaxial	10 Mbps (often 4-6 Mbps in practice)	Cheaper, thinner cabling than 10Base5
10Base-T	Twisted pair (Cat 3/Cat 5)	10 Mbps	Introduced the RJ45/star-topology model still used today
10Base-F	Multimode fibre	10 Mbps, up to 2 km/segment	ST or SMA connectors, star topology via fibre hub
Fast Ethernet (802.3u)	Cat5 (TX) or fibre (FX)	100 Mbps	Backward compatible with 10Base-T
Gigabit Ethernet (802.3z/802.3ab)	Cat5e/6 (T) or fibre (SX/LX)	1 Gbps	Full-duplex; standard for data centres and backbones

8.5 Media Access: CSMA/CD vs. CSMA/CA

Whenever multiple devices share one transmission medium, there needs to be a rule for who gets to "speak" and what happens when two devices speak at once.

Aspect	CSMA/CD	CSMA/CA
Full name	Carrier Sense Multiple Access / Collision Detection	Carrier Sense Multiple Access / Collision Avoidance
Strategy	Listen, transmit, and detect collisions <i>after</i> they happen	Signal intent to transmit and wait for acknowledgment <i>before</i> sending, to avoid collisions altogether
Network type	Wired networks	Wireless networks (radio can't reliably "hear" its own collision)
IEEE standard	802.3	802.11

8.6 Baseband vs. Broadband

BASEBAND

Uses the entire bandwidth of the medium for one digital signal at a time; simple and common in LANs (e.g. 10Base-T, 100Base-T); needs repeaters for long-distance transmission.

BROADBAND

Divides the medium's bandwidth into multiple frequency channels, carrying several analogue signals simultaneously; suited to long-distance links such as DSL and cable internet.

Chapter Summary

- IEEE Project 802 standardised LAN/MAN technologies so equipment from different vendors could interoperate.
- Ethernet (802.3), Token Bus (802.4), and Token Ring (802.5) each solved shared-medium access differently; Ethernet's simplicity is why it dominates today.
- FDDI extended token-passing to fibre optic backbones spanning up to 200 km.
- Ethernet evolved from 10 Mbps coax (10Base5/2) through twisted-pair and fibre (10Base-T/F) to Fast (100 Mbps) and Gigabit (1 Gbps) Ethernet.
- CSMA/CD detects collisions after they happen (wired); CSMA/CA avoids them beforehand (wireless).

Check Yourself

1. Why does a wireless network need CSMA/CA rather than CSMA/CD?
2. What key structural difference distinguishes Token Bus from Token Ring, despite both using token passing?
3. Which Ethernet generation introduced the star-topology, RJ45-based wiring model still in use today?
4. Why is FDDI's dual-ring design considered a reliability feature rather than just added complexity?



CHAPTER 9

Wide Area Networks

Connecting cities and continents means giving up the luxury of a single dedicated wire for every conversation.

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- Contrast circuit switching and packet switching
- Compare leased lines, ISDN, and DSL as WAN connection options
- Distinguish ADSL from SDSL

9.1 Circuit Switching vs. Packet Switching

A WAN has to move data over long distances using shared, expensive infrastructure. There are two fundamentally different philosophies for doing this.

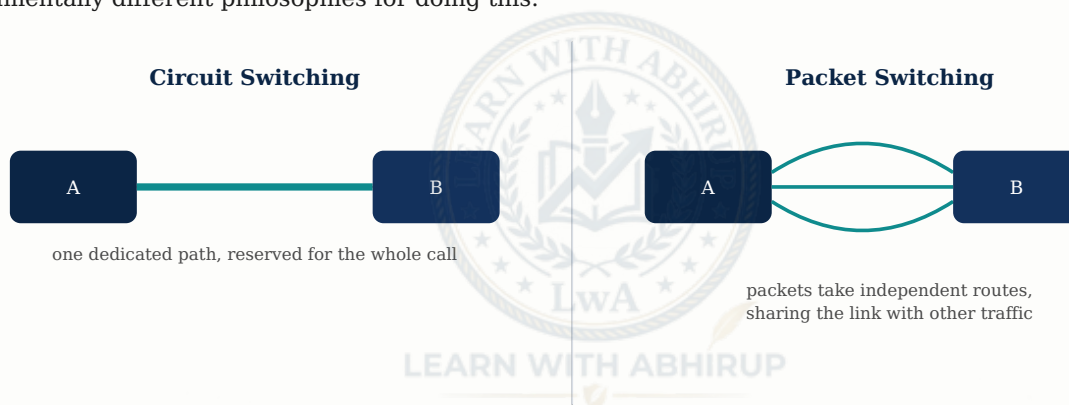


Figure 9.1 — Circuit switching reserves one dedicated path for an entire session; packet switching lets independent packets share the link, taking whichever route is available.

	Circuit Switching	Packet Switching
Connection	Dedicated path reserved for the whole session	Data split into packets that travel independently
Bandwidth guarantee	Yes — reserved regardless of use	No — shared, more efficient use of available capacity
Analogy	A traditional phone call	Postal mail — many letters share the same roads
Examples	ISDN, PSTN (plain old telephone system)	DSL, ADSL, the internet itself

9.2 WAN Connection Technologies

Leased Line

A dedicated, always-on, high-bandwidth connection between two fixed locations. It is the most reliable and consistent option available — and the most expensive — making it the standard choice for critical business links that cannot tolerate downtime.

ISDN (Integrated Services Digital Network)

Carries digital voice, data, and video over existing phone lines. Faster and more reliable than analogue phone lines, but limited to roughly 64 kbps per channel — a ceiling that broadband technologies quickly outgrew, leaving ISDN largely superseded today.

DSL (Digital Subscriber Line)

A family of technologies that reuse existing copper telephone lines to deliver speeds of several Mbps — far beyond ISDN — while usually leaving the line free for simultaneous voice calls.

ADSL (ASYMMETRIC)

Higher download speed than upload speed — ideal for typical home/internet usage, where downloading (web pages, video) dominates.

SDSL (SYMMETRIC)

Equal upload and download speeds — better suited to business applications like hosting servers or video conferencing, where upload matters as much as download.

CHOOSING A WAN TECHNOLOGY

The right choice depends on bandwidth needs, budget, and reliability requirements. Leased lines dominate where guaranteed uptime is essential; packet-switched technologies like DSL dominate general internet access because they are far more cost-efficient for bursty, everyday traffic.

Chapter Summary

- Circuit switching reserves a dedicated path for an entire session (ISDN, PSTN); packet switching shares the link efficiently among many independent transmissions (DSL, the internet).
- Leased lines offer the highest reliability at the highest cost; ISDN has largely been superseded by broadband; DSL is the dominant technology for cost-effective, high-speed internet access.
- ADSL favours download-heavy home use; SDSL favours upload-balanced business use.

Check Yourself

1. Why is a traditional phone call a good analogy for circuit switching specifically?
2. A small business needs to host its own web server with heavy two-way traffic. Would ADSL or SDSL suit it better, and why?
3. Why has packet switching become the dominant approach for general internet access, despite offering no bandwidth guarantee?

CHAPTER 10

Wireless Networking and Security

Cutting the cable comes with trade-offs in speed, range, and security — and a rulebook of its own.

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

- Name the major wireless standards and their frequency bands
- Compare Bluetooth, HomeRF, and IrDA
- Distinguish ad-hoc from infrastructure wireless networks
- List practical steps to secure a Wi-Fi network

10.1 How Wireless Networking Works

Wireless networking transmits data as electromagnetic waves through the air instead of down a cable. This trades the reliability and shielding of physical media for mobility — at the cost of being more exposed to interference and eavesdropping.

Key wireless standards

Standard	Purpose
IEEE 802.11 (Wi-Fi)	Wireless LANs; variants a/b/g/n/ac/ax
Bluetooth	Short-range device-to-device connections
Zigbee	Low-power, low-data-rate — home automation and IoT
WiMAX (IEEE 802.16)	High-speed broadband wireless over long distances
LTE / 5G	Cellular technologies for mobile data

802.11 Wi-Fi generations at a glance

Standard	Frequency	Max Speed	Typical Indoor Range
802.11	2.4 GHz	1–2 Mbps	~20 feet
802.11a	5 GHz	54 Mbps	25–75 feet
802.11b	2.4 GHz	11 Mbps	Up to 150 feet
802.11g	2.4 GHz	54 Mbps	Up to 150 feet
802.11n	2.4/5 GHz	600 Mbps	175+ feet

All of these use **CSMA/CA** (Chapter 8) as their access method, since a wireless station cannot reliably detect a collision while it is transmitting.

10.2 Short-Range Wireless Technologies

Technology	Range	Data Rate	Line of Sight?	Typical Use
Bluetooth	~10 m	~1 Mbps+	No	Headsets, speakers, peripherals, piconets of up to 8 devices
HomeRF	~50 m	Up to 10 Mbps	No	Home area networks (largely displaced by Wi-Fi)
IrDA	~1 m	115 kbps–4 Mbps	Yes — required	Legacy point-to-point transfer between laptops/PDAs/printers

10.3 Ad-Hoc vs. Infrastructure Networks

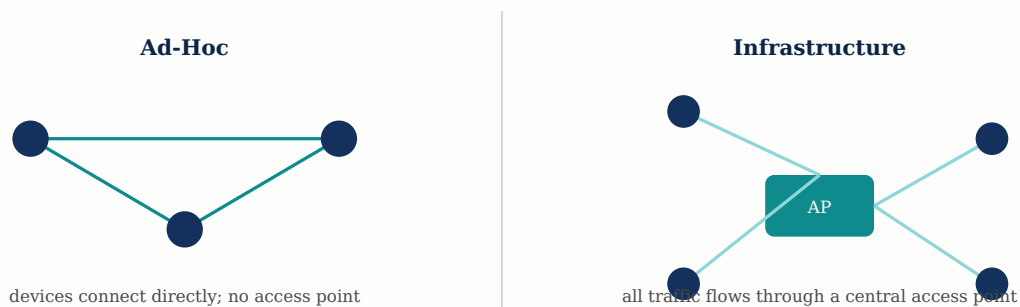


Figure 10.1 — An ad-hoc network has no central coordinator; an infrastructure network routes everything through an access point.

	Ad-Hoc	Infrastructure
Structure	Decentralised, peer-to-peer	Centralised around an access point (AP)
Setup	Quick, minimal configuration	Requires configuring the AP
Typical scale	2–4 devices, temporary	Homes, offices, public hotspots
Security	Manual, per-device configuration	Centrally managed, generally stronger

10.4 Wi-Fi Security Protocols

Protocol	Security level
WEP	Legacy, weak encryption — should not be used today
WPA	Improved over WEP, still eventually superseded
WPA2	Strong — uses AES encryption; long-standing standard
WPA3	Current standard, strongest available protections

SECURING A WIRELESS NETWORK — A PRACTICAL CHECKLIST

- **Strong password + WPA2/WPA3** — replace default router credentials; enable the strongest encryption your hardware supports.
- **SSID hygiene** — avoid personal or identifying network names; consider hiding SSID broadcast.
- **Guest network** — isolate visitor devices from your main network.
- **Firmware updates** — patch known router vulnerabilities regularly.
- **Device management** — limit and audit which devices are allowed to connect.
- **Firewall enabled** — filter traffic at the router.
- **VPN on public Wi-Fi** — encrypt your traffic whenever you're on a network you don't control.

Chapter Summary

- Wireless networking trades cable reliability for mobility, using standards like 802.11 (Wi-Fi), Bluetooth, Zigbee, and WiMAX across different frequency bands.
- Bluetooth, HomeRF, and IrDA occupy different niches by range, speed, and whether they need line-of-sight — IrDA's requirement for line-of-sight is why it has largely disappeared.
- Ad-hoc networks are decentralised and temporary; infrastructure networks route through a central access point and scale far better.
- WPA2/WPA3 with a strong password, an isolated guest network, and up-to-date firmware form the baseline of a secure home or office Wi-Fi setup.

Check Yourself

1. Why do all 802.11 Wi-Fi variants use CSMA/CA rather than CSMA/CD?
2. A user wants to transfer a file between two laptops sitting side by side, with no network infrastructure available. Which technology from Section 10.2 is most appropriate, and why?
3. Why is WEP no longer considered acceptable for securing a wireless network?
4. Explain, using the ad-hoc/infrastructure distinction, why a coffee shop's public Wi-Fi is necessarily an infrastructure network rather than ad-hoc.

GLOSSARY

Key Terms A-Z

Every bolded term from the book, in one place.

Term	Definition
ADSL	Asymmetric DSL — faster download than upload speed; suited to typical home use.
APIPA	Automatic Private IP Addressing (169.254.x.x) — self-assigned when a device cannot reach a DHCP server.
ARP	Address Resolution Protocol — maps a device's IP address to its MAC address on a local network.
Bridge	A Layer 2 device that joins two network segments, filtering traffic by MAC address.
Broadcast address	An address that delivers data to every device on a network segment at once.
CIDR	Classless Inter-Domain Routing — notation expressing a subnet mask as a slash and a bit count (e.g. /24).
Circuit switching	A WAN method that reserves one dedicated path for the entire duration of a session.
Client-server	An architecture where dedicated servers provide centralised services to many client devices.
CSMA/CA	Carrier Sense Multiple Access / Collision Avoidance — the wireless media-access method used by 802.11.
CSMA/CD	Carrier Sense Multiple Access / Collision Detection — the wired media-access method used by Ethernet (802.3).
DSL	Digital Subscriber Line — high-speed data over existing telephone copper wiring.
FDDI	Fiber Distributed Data Interface — a 100 Mbps, dual-ring, fibre-based LAN standard spanning up to 200 km.
Firewall	A device or program that filters network traffic according to a defined security policy.
Gateway	A device that translates between two dissimilar networks or protocol systems.
Hub	A Layer 1 device that broadcasts every incoming signal to all connected ports.
IEEE 802	The family of standards governing LAN and MAN technologies.
IP address	A numeric label (32 bits in IPv4) uniquely identifying a device on a network.
ISDN	Integrated Services Digital Network — digital voice/data/video over phone lines, limited to ~64 kbps.
MAC address	A permanent, hardware-based identifier burned into a NIC by its manufacturer.
Multicast	Delivery from one sender to a defined group of recipients (224.0.0.0–239.255.255.255).
NAT	Network Address Translation — lets many private addresses share one public address.
NIC	Network Interface Card — the hardware providing a device's physical network connection.
Node	Any device attached to a network capable of sending, receiving, or forwarding data.
OSI Model	A seven-layer conceptual framework (ISO) describing network communication functions.

Packet	The fundamental unit of data carried over a network, with header/trailer control information.
Packet switching	A WAN method where data is split into independently routed packets sharing the link.
PAT	Port Address Translation — a NAT variant sharing one public address among many devices via port numbers.
Peer-to-peer	An architecture where every node has equal standing, with no dedicated central server.
Private IP address	An address reserved for internal network use, never routed on the public internet.
Protocol	An agreed set of rules governing how data is formatted, transmitted, and interpreted.
Public IP address	A globally unique address, allocated by IANA/RIRs, routable on the public internet.
Router	A Layer 3 device that forwards IP packets between different networks.
SAN	Storage Area Network — a dedicated, high-speed network connecting storage devices to servers.
SDSL	Symmetric DSL — equal upload and download speeds; suited to business use.
Subnet mask	A 32-bit pattern marking the boundary between an address's network and host portions.
Subnetting	Borrowing host bits to divide one network into multiple smaller subnetworks.
Switch	A Layer 2 device that forwards frames only to the port of the intended recipient.
TCP/IP Model	The four-layer protocol suite (Network Access, Internet, Transport, Application) that actually runs the internet.
Token Ring / Token Bus	LAN standards (IEEE 802.5 / 802.4) using a circulating token to control which station may transmit.
Topology	The arrangement of nodes and links in a network — physical (visible cabling) or logical (data path).
Unicast	Delivery from one sender to exactly one specific recipient.
VPN	Virtual Private Network — an encrypted tunnel across a public network for secure remote access.
WPA2 / WPA3	Wi-Fi Protected Access security protocols; WPA2 uses AES encryption, WPA3 is the current strongest standard.

LEARN WITH ABHIRUP

Question Bank

Multiple-choice, short-answer, and long-answer questions organised by chapter, for revision and self-testing. An answer key for every MCQ follows at the end.

Chapter 1 — What Is a Computer Network?

MCQs

1. Which of the following is *not* one of the four core components of a network exchange?
A) Sender B) Receiver C) Protocol D) Firewall
2. A network confined to a single office building is best classified as a:
A) WAN B) LAN C) MAN D) SAN
3. Which network type is specifically dedicated to connecting storage devices to servers?
A) VPN B) CAN C) SAN D) PAN
4. A VPN is best described as a:
A) Geographic network category B) Security technique creating an encrypted tunnel C) Type of topology D) Type of NIC
5. Which of the following covers the largest geographic area?
A) PAN B) LAN C) CAN D) WAN

Short Answer (3 marks)

1. Explain why resource sharing is considered a primary motivation for building networks.
2. Differentiate a CAN from a MAN in terms of ownership and management.

Long Answer (5 marks)

1. Describe the four components present in every network data exchange, and explain the role each plays using a real-world example (e.g. sending an email).

Chapter 2 — Network Architectures and Topologies

MCQs

1. In a peer-to-peer network, each node can act as:
A) Only a client B) Only a server C) Both client and server D) Neither
2. Which topology is most vulnerable to a single point of failure at its central device?
A) Bus B) Star C) Mesh D) Ring
3. Which topology offers the highest fault tolerance through redundant paths?
A) Bus B) Ring C) Mesh D) Star
4. The actual visible layout of cables and devices in a network is called its:
A) Logical topology B) Physical topology C) Virtual topology D) Network model
5. Most large real-world organisational networks are best described as:
A) Pure bus B) Pure ring C) Hybrid D) Pure mesh

Short Answer (3 marks)

1. Explain the difference between physical and logical topology with an example.
2. Why is client-server architecture generally preferred over peer-to-peer in large organisations?

Long Answer (5 marks)

1. Compare bus, star, ring, and mesh topologies on cost, fault tolerance, and scalability, and recommend one for a hospital's critical monitoring network with justification.

Chapter 3 — The Internet and Its Applications

MCQs

1. The internet is best described as a:
A) Single centrally-owned network B) Network of networks C) Type of LAN D) Type of topology
2. Which entity provides the physical infrastructure connecting a home network to the internet backbone?
A) NIC B) ISP C) Firewall D) Switch
3. Which protocol suite underlies almost all internet communication?
A) IEEE 802 B) TCP/IP C) CSMA/CA D) WPA2
4. Which protocol is used to *send* email?
A) IMAP B) POP3 C) SMTP D) FTP
5. Streaming media and online gaming fall under which category of network application?
A) Communication B) File sharing C) Online services D) Security

Short Answer (3 marks)

1. List, in order, the components a request typically passes through from your device to a destination server.
2. Distinguish SMTP from POP3/IMAP in the email delivery process.

Long Answer (5 marks)

1. Explain why the internet's design as a "network of networks" makes universal addressing and shared protocols essential, rather than optional.

Chapter 4 — Network Devices

MCQs

1. A hub operates at which OSI layer?
A) 1 B) 2 C) 3 D) 4
2. Which device forwards data based on MAC addresses?
A) Hub B) Switch C) Modem D) Repeater
3. Which device connects different networks together using IP addresses and a routing table?
A) Bridge B) Router C) Hub D) NIC
4. A MAC address's first six hexadecimal digits identify the:
A) Device's IP subnet B) Manufacturer (OUI) C) Port number D) Protocol version
5. Which firewall type inspects traffic at the application layer, hiding the client's real address?
A) Packet filtering B) Proxy firewall C) Hardware firewall D) Cloud firewall

Short Answer (3 marks)

1. Explain why switches reduce network congestion compared to hubs.
2. What is ARP, and why is it necessary alongside MAC and IP addresses?

Long Answer (5 marks)

1. Compare routers, bridges, and gateways in terms of OSI layer and primary function, giving one scenario

where each would be the correct device to deploy.

Chapter 5 — Transmission Media and Connectors

MCQs

1. Which cable type is immune to electromagnetic interference?
A) Coaxial B) UTP C) STP D) Fibre optic
2. Which connector is standard for Ethernet networking?
A) RJ11 B) RJ45 C) BNC D) F-Type
3. Which twisted-pair category supports up to 10 Gbps at 500 MHz?
A) Cat 5e B) Cat 6 C) Cat 6a D) Cat 3
4. Single-mode fibre is best suited for:
A) Short, cheap in-building runs B) Long-distance, high-bandwidth links C) Telephone lines only D) Cable television only
5. Which coaxial connector is known for a bayonet locking mechanism used in professional video?
A) F-Type B) N-Type C) BNC D) SC

Short Answer (3 marks)

1. Explain the practical difference between RJ45 and RJ11 connectors.
2. Why is multi-mode fibre generally cheaper than single-mode fibre, and when is that trade-off acceptable?

Long Answer (5 marks)

1. A company is wiring a new office and a separate warehouse 30 km away. Recommend cabling choices for each link and justify them using the properties covered in this chapter.

Chapter 6 — Reference Models: OSI and TCP/IP

MCQs

1. Which OSI layer handles encryption and data formatting?
A) Session B) Presentation C) Transport D) Application
2. TCP and UDP operate at which OSI layer?
A) 2 B) 3 C) 4 D) 7
3. The TCP/IP model's "Network Access" layer corresponds to which OSI layers?
A) 7 and 6 B) 5 and 4 C) 3 only D) 2 and 1
4. Who developed the OSI model?
A) ARPANET B) ISO C) IEEE D) IANA
5. How many layers does the TCP/IP model have?
A) 4 B) 5 C) 7 D) 3

Short Answer (3 marks)

1. List the seven OSI layers in order from Application to Physical.
2. Explain one practical benefit of using a layered model when troubleshooting a network fault.

Long Answer (5 marks)

1. Map all seven OSI layers onto the four TCP/IP layers, explaining which OSI layers merge into each TCP/IP layer and why.

Chapter 7 — IP Addressing, NAT, and Subnetting

MCQs

1. Which address class is reserved for multicast groups?

- A) A B) B C) C D) D
2. Which of the following is a private IP address range?
A) 8.8.8.0/24 B) 192.168.0.0/16 C) 172.32.0.0/16 D) 203.0.113.0/24
3. What does CIDR notation "/24" indicate?
A) 24 usable hosts B) 24 leading 1-bits in the subnet mask C) 24 subnets D) Class C by default only
4. An address in the 169.254.x.x range indicates:
A) A public internet address B) A multicast address C) APIPA self-assignment (no DHCP reached) D) A loopback address
5. PAT allows multiple devices to share one public address by distinguishing them using:
A) MAC addresses B) Port numbers C) Subnet masks D) VLAN tags

Short Answer (3 marks)

1. Explain the difference between a network address and a broadcast address within a subnet.
2. Why must you know an address's default class mask before calculating how many bits were borrowed for subnetting?

Long Answer (5 marks)

1. Given the network 10.10.0.0/16 subnetted to /21, calculate the number of subnets created and the number of usable hosts per subnet, showing all working.

Chapter 8 — LAN Technologies and Ethernet

MCQs

1. IEEE 802.3 defines the standard for:
A) Token Ring B) Token Bus C) Ethernet D) Wi-Fi
2. Which access method does wired Ethernet use?
A) CSMA/CA B) CSMA/CD C) Token passing D) TDMA
3. FDDI can span distances up to:
A) 500 m B) 2 km C) 200 km D) Unlimited
4. Which Ethernet standard first introduced twisted-pair, star-topology wiring?
A) 10Base5 B) 10Base2 C) 10Base-T D) 10Base-F
5. Gigabit Ethernet over Cat5e/Cat6 cable is designated:
A) 100Base-TX B) 1000Base-T C) 10Base-T D) 1000Base-SX

Short Answer (3 marks)

1. Explain how the token-passing mechanism prevents collisions in Token Ring.
2. Distinguish baseband from broadband transmission with one example of each.

Long Answer (5 marks)

1. Compare IEEE 802.3, 802.4, and 802.5 on topology, priority support, and suitability for real-time traffic, and explain why Ethernet still became the dominant LAN standard despite not scoring best on every criterion.

Chapter 9 — Wide Area Networks

MCQs

1. Which switching method reserves a dedicated path for the whole session?
A) Packet switching B) Circuit switching C) Cell switching D) Label switching
2. Which WAN option offers the highest reliability at the highest cost?
A) DSL B) ISDN C) Leased line D) Dial-up
3. ADSL is best suited to:
A) Equal upload/download needs B) Download-heavy home use C) Only voice calls D) Only LAN

traffic

4. ISDN has largely been superseded because of its:
A) High cost only B) Limited speed (~64 kbps) C) Lack of digital signalling D) Incompatibility with phone lines
5. Packet switching is dominant for internet access mainly because it is:
A) More secure B) More bandwidth-efficient for bursty traffic C) Guaranteed bandwidth D) Cheaper hardware only

Short Answer (3 marks)

1. Explain why a leased line, despite its cost, remains preferred for certain business applications.
2. Differentiate ADSL from SDSL and give one appropriate use case for each.

Long Answer (5 marks)

1. Compare circuit switching and packet switching in terms of reliability, efficiency, and cost, and explain which approach the modern internet relies on and why.

Chapter 10 — Wireless Networking and Security

MCQs

1. Which frequency band is most commonly used by both Wi-Fi and Bluetooth?
A) 900 MHz B) 2.4 GHz C) 5 GHz D) 60 GHz
2. IrDA requires which of the following to function?
A) A central access point B) Line-of-sight between devices C) A wired backbone D) A SIM card
3. Which Wi-Fi security protocol currently offers the strongest protection?
A) WEP B) WPA C) WPA2 D) WPA3
4. An ad-hoc wireless network is best described as:
A) Centralised around an access point B) Decentralised, direct device-to-device C) Always encrypted by default D) Limited to cellular networks
5. Which technology directs a wireless signal specifically toward one device to improve connection quality?
A) MIMO B) OFDM C) Beamforming D) CSMA/CA

Short Answer (3 marks)

1. List three practical steps to secure a home Wi-Fi network.
2. Explain why Bluetooth is preferred over IrDA for most modern short-range applications.

Long Answer (5 marks)

1. Compare ad-hoc and infrastructure wireless networks on structure, security, and scalability, and recommend which is appropriate for a public library offering Wi-Fi to patrons.

Answer Key — All MCQs

Ch.	Answers (Q1-Q5)				
1	D	B	C	B	D
2	C	B	C	B	C
3	B	B	B	C	C
4	A	B	B	B	B
5	D	B	C	B	C
6	B	C	D	B	A
7	D	B	B	C	B
8	C	B	C	C	B
9	B	C	B	B	B
10	B	B	D	B	C

Short-answer and long-answer questions are designed for written practice and instructor evaluation; model answers can be constructed directly from the corresponding chapter section referenced in each question.

